

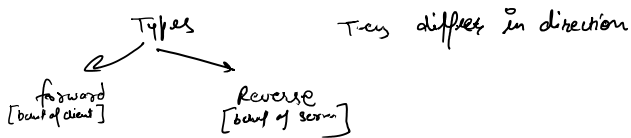
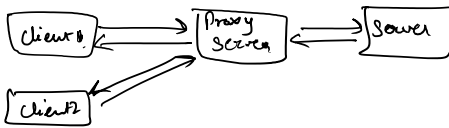
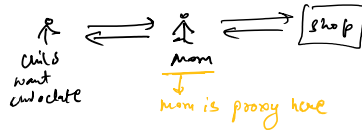
Proxy vs reverse proxy

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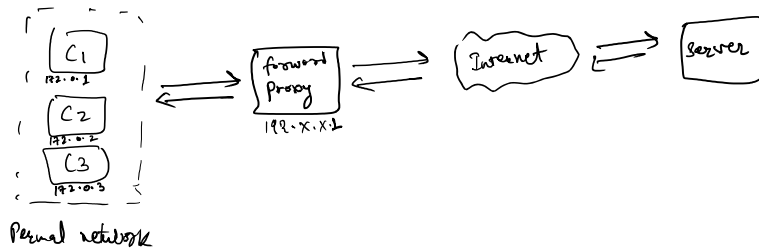
Proxy VS Reverse Proxy

→ What is Proxy Service?

Proxy is taking request on someone behalf



Forward proxy :- Normally when we talk about proxy we are talking about forward proxy



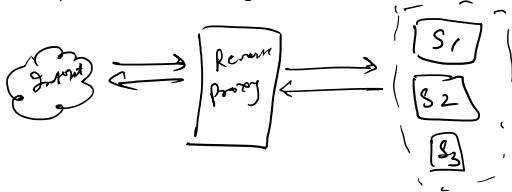
- It hides the client network from outside world so it gives Anonymousity to inner networks
- On the outer world proxy sends its IP address instead of client
- When C₂ sends request having IP 172.0.1 proxy will change IP 172.X.X.1 to server and return back. So server will not know request is coming from which client
- Grouping of Request :- It can club the request and sends to server.
ex. C₁, C₂, C₃ calls google.com so proxy will club those request and send to server in single request
- Help Access Restricted Data/Content :- You can access restricted data
ex. In India PubG server cannot be accessed so if we use proxy which is in US then PUBG server doesn't know the request is coming from India. It will think request is coming from US
- Security :- You can add rules to access the content or you can control request access
ex. No social media so proxy will not return response from any social media website
- Caching :- Proxy cache the data helps to reduce unnecessary calls to server
ex. C₂ wants some data, proxy checks in cache if it does not present then it send request to server and save response in cache and after that if any other client request same data then immediately it will return.
Mostly cache data is static data

Disadvantage :-

1) Application Level :- if you have 10 Application than you have to set proxy for all 10 App

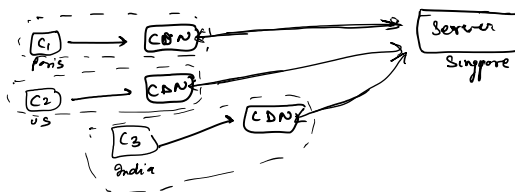
Reverse Proxy ex: CDN

No Request can directly talk to Server



Advantage

1) Security :- outer world doesnot know about IP of Servers. they only knows about IP of reverse Proxy. So if Hacker tries DDos Attack it can attack on reverse Proxy, not on servers

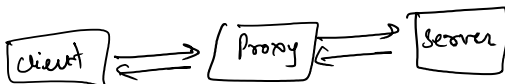


2) Latency :- It reduce the latency by caching

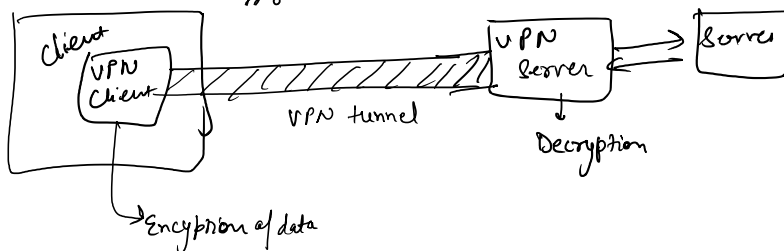
3) Loadbalancer Capability :- It can re-direct the request to different servers
So it can help in Load balancing

Proxy vs VPN

→ Major difference is VPN can do encryption and decryption and it creates tunnel to transfer the data safely and proxy do not do



anonymity (hide IP)
Caching
Logging



Proxy vs Load balancer

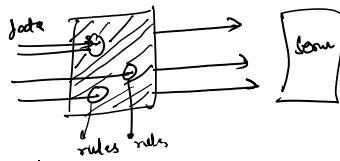
→ Reverse Proxy can act as loadbalancer but LB cannot be a proxy

→ LB cannot do caching, Logging and IP hiding

→ LB is required when there are more than 1 server but RP might be required

Proxy vs Firewall

Firewall can do packet scanning



- Each host has rules to allow data.
- Packet scanning means it will check header, Port No, IP of source & destination
- In proxy you can set rules on data also and it sets up for application level
- Proxy can act as firewall also as we can block and allow requests and some times this feature called proxy firewall and it work on application layer